

# **Lab Assignment 9**

## **Object-Oriented Programming**

**Branch**

**B.E., 2<sup>nd</sup> Year – ENC**



**Submitted By:**

**Ryan kapoor**

**Roll No. 1024150188**

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**1) Write a C++ program to write numbers 1 to 200 in a data file NUM TXT.**

**Input:**

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
int main() {
```

```
    ofstream file("NUM.txt");
```

```
    for(int i = 1; i <= 200; i++) {
```

```
        file << i << " ";
```

```
    }
```

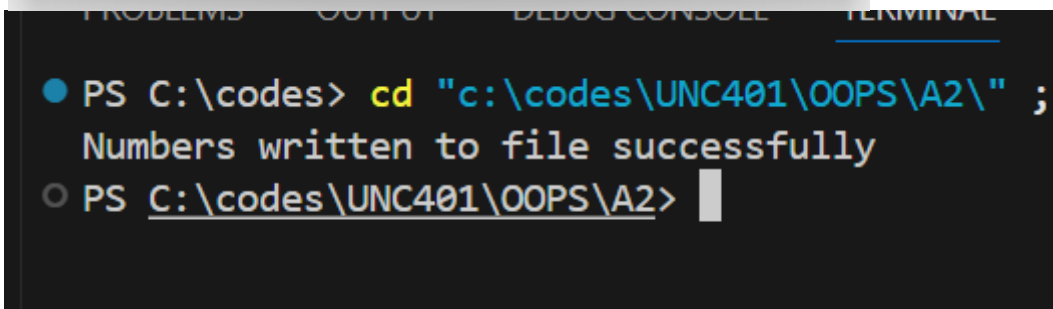
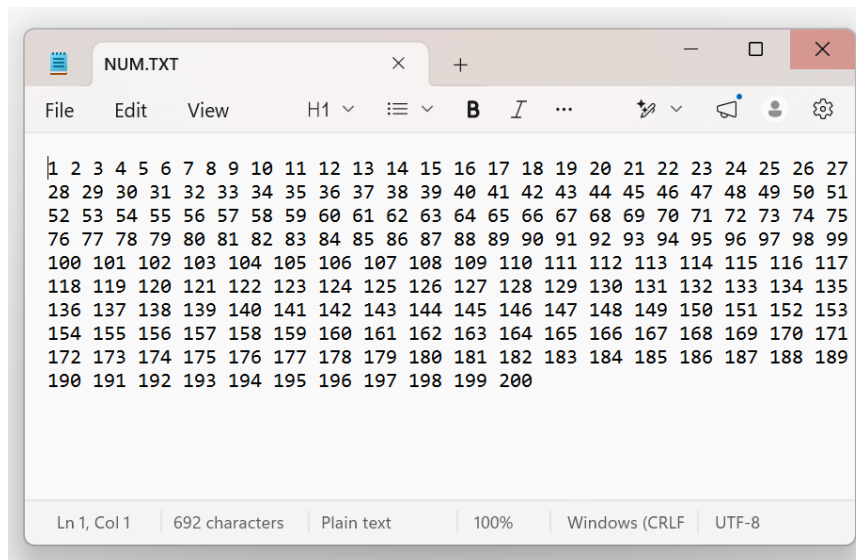
```
    file.close();
```

```
    cout << "Numbers written to file successfully" << endl;
```

```
    return 0;
```

```
}
```

## Output:



2) Write a user-defined function in C++ to read the content from a text file NOTES.TXT, count and display the number of alphabets present in it.

### Input:

```
#include <iostream>
```

```
#include <fstream>
```

```
#include <cctype>
```

```
using namespace std;
```

```
void countAlphabets() {  
    ifstream file("NOTES.TXT");  
    char ch;  
    int count = 0;  
  
    if(!file) {  
        cout << "File not found" << endl;  
        return;  
    }  
}
```

```
while(file.get(ch)) {  
    if(isalpha(ch)) {
```

```
        count++;
    }
}

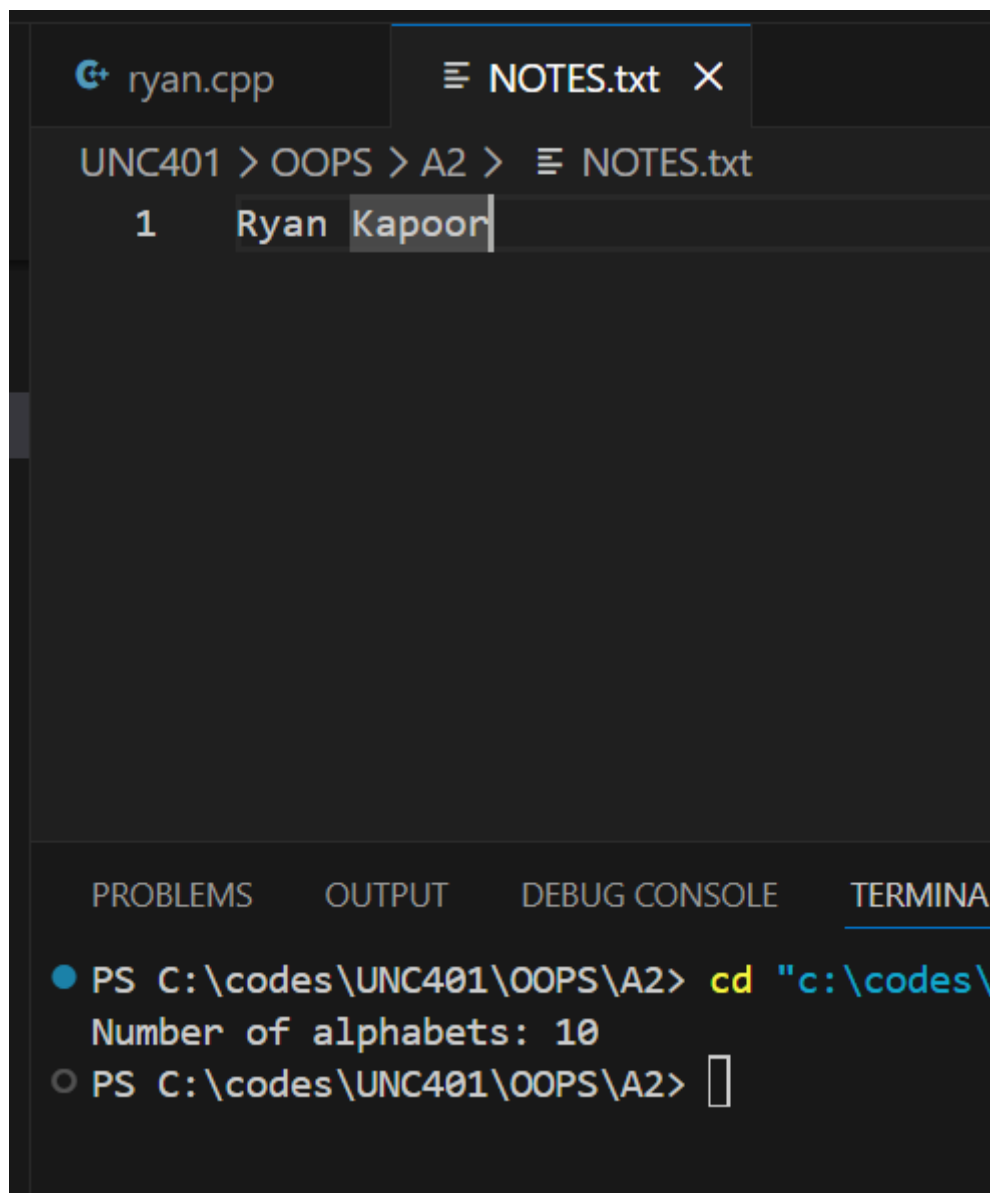
file.close();

cout << "Number of alphabets: " << count << endl;
}

int main() {

    countAlphabets();

    return 0;
}
```



The screenshot shows a code editor with two tabs: 'ryan.cpp' and 'NOTES.txt'. The 'ryan.cpp' tab is active, displaying the C++ code from the previous block. Below the code editor, there is a terminal window. The terminal shows the command prompt 'UNC401 > OOPS > A2 >' followed by the command 'cd "c:\codes\' and the output 'Number of alphabets: 10'. The terminal also shows the command prompt 'PS C:\codes\UNC401\OOPS\A2>' followed by a blank line.

```
UNC401 > OOPS > A2 > cd "c:\codes\
Number of alphabets: 10
PS C:\codes\UNC401\OOPS\A2>
```

**Output:**

**3) Write a program to copy the contents of one file to another.**

**Input:**

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
int main() {
```

```
    ifstream src("source.txt");
```

```
    ofstream dest("dest.txt");
```

```
    char ch;
```

```
    while (src.get(ch)) {
```

```
        dest.put(ch);
```

```
    }
```

```
cout << "File copied successfully";

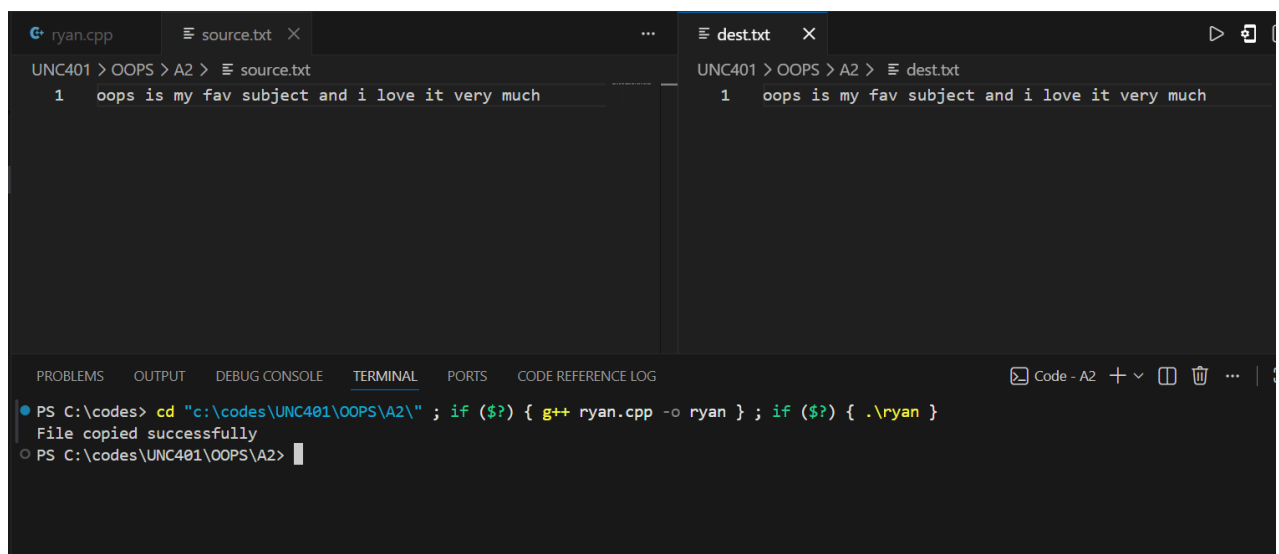
src.close();

dest.close();

return 0;

}
```

### Output:



The screenshot shows a Visual Studio Code editor with two source files open: `ryan.cpp` and `dest.txt`. Both files contain the same text: `1 oops is my fav subject and i love it very much`. The terminal at the bottom shows the command `cd "c:\codes\UNC401\OOPS\A2\" ; if ($?) { g++ ryan.cpp -o ryan } ; if ($?) { .\ryan }` being executed, with the output `File copied successfully`.

```
ryan.cpp  source.txt  dest.txt
UNC401 > OOPS > A2 > source.txt
1 oops is my fav subject and i love it very much
UNC401 > OOPS > A2 > dest.txt
1 oops is my fav subject and i love it very much

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  CODE REFERENCE LOG
PS C:\codes> cd "c:\codes\UNC401\OOPS\A2\" ; if ($?) { g++ ryan.cpp -o ryan } ; if ($?) { .\ryan }
File copied successfully
PS C:\codes\UNC401\OOPS\A2>
```

**4) Write a program to implement I/O operations on characters. I/O operations include inputting a string, calculating the length of the string, storing the string in a file and fetching the stored characters from it.**

**Input:**

```
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;

int main() {

    char str[100];

    cout << "Enter a string: ";
    cin.getline(str, 100);

    int len = strlen(str);
    cout << "Length of string: " << len << endl;

    ofstream fout("DATA.txt");
    fout << str;
    fout.close();

    ifstream fin("DATA.txt");
    char ch;

    cout << "Stored characters from file: ";
    while(fin.get(ch)) {
        cout << ch;
    }

    fin.close();

    return 0;
}
```

**Output:**

UNC401 > OOPS > A2 > DATA.txt

1   oops is very good

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

CODE RE

PS C:\codes\UNC401\OOPS\A2>

- `cd "c:\codes\UNC401\OOPS\A2\"`  
Enter a string: oops is very good  
Length of string: 17  
Stored characters from file: oops is very good
- PS C:\codes\UNC401\OOPS\A2>



5) Write a program to do the followings:

(a) Create a file A-Z and read 10<sup>th</sup> character using seekg().

(b) Overwrite 5<sup>th</sup> number using seekp().

(c) Find file size using tellg().

(d) Read last character using seekg().

(e) Create a file data.txt with multiple lines. Read the file and:

- i. Move the pointer to the 10<sup>th</sup> byte using seekg()
- ii. Display the current position using tellg().
- iii. Read and display the remaining content.

**Input:**

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {

    fstream file;

    file.open("A-Z.txt", ios::out);
    for(char ch = 'A'; ch <= 'Z'; ch++) {
        file << ch;
    }
    file.close();

    file.open("A-Z.txt", ios::in);
    file.seekg(9);
    char ch;
    file.get(ch);
    cout << "10th character: " << ch << endl;
    file.close();

    file.open("A-Z.txt", ios::in | ios::out);
    file.seekp(4);
    file.put('X');
    file.close();

    file.open("A-Z.txt", ios::in);
    file.seekg(0, ios::end);
    cout << "File size: " << file.tellg() << " bytes" << endl;
    file.close();

    file.open("A-Z.txt", ios::in);
    file.seekg(-1, ios::end);
    file.get(ch);
    cout << "Last character: " << ch << endl;
    file.close();
```

```
file.open("data.txt", ios::out);
file << "This is line 1\n";
file << "This is line 2\n";
file << "This is line 3\n";
file.close();

file.open("data.txt", ios::in);
file.seekg(10);
cout << "Current position: " << file.tellg() << endl;

cout << "Remaining content:\n";
while(file.get(ch)) {
    cout << ch;
}
file.close();

return 0;
}
```

**Output:**

UNC401 > OOPS > A2 > ≡ A-Z.txt

1 ABCDXFGHIJKLMNOPQRSTUVWXYZ

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PS C:\codes\UNC401\OOPS\A2>

cd "c:\codes\UN

10th character: J

File size: 26 bytes

Last character: Z

Current position: 10

Remaining content:

ne 1

This is line 2

This is line 3

PS C:\codes\UNC401\OOPS\A2>

UNC401 > OOPS > A2 > DATA.txt

```
1 This is line 1
2 This is line 2
3 This is line 3
4
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

PS C:\codes\UNC401\OOPS\A2>

cd "c:\cod

10th character: J  
File size: 26 bytes  
Last character: Z  
Current position: 10  
Remaining content:  
ne 1  
This is line 2  
This is line 3

PS C:\codes\UNC401\OOPS\A2>

6) Write a program to do the followings:

- i. Create a file with text "Hello World"
- ii. After writing each character, display the current position of the put pointer using tellp().
- iii. Replace "World" with "C++" using random access

**Input:**

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {

    fstream file;

    file.open("text.txt", ios::out);

    string str = "Hello World";

    for(int i = 0; i < str.length(); i++) {
        file.put(str[i]);
        cout << "Position after writing '" << str[i] << "': " << file.tellp() << endl;
    }

    file.close();

    file.open("text.txt", ios::in | ios::out);

    file.seekp(6);

    file.put('C');
    file.put('+');
    file.put('+');
    file.put(' ');
    file.put(' ');

    file.close();

    file.open("text.txt", ios::in);
    char ch;

    cout << "Final content: ";
    while(file.get(ch)) {
        cout << ch;
    }

    file.close();

    return 0;
}
```

**Output:**

```
cd "c:\codes\  
Position after writing 'H': 1  
Position after writing 'e': 2  
Position after writing 'l': 3  
Position after writing 'l': 4  
Position after writing 'o': 5  
Position after writing ' ': 6  
Position after writing 'W': 7  
Position after writing 'o': 8  
Position after writing 'r': 9  
Position after writing 'l': 10  
Position after writing 'd': 11  
Final content: Hello C++  
PS C:\codes\UNC401\OOPS\A2>
```

```
ryan.cpp  text.txt  
UNC401 > OOPS > A2 > text.txt  
1 Hello C++
```